Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/08/2025
PI Name:	Smith, Amanda L. Ph.D.		
Project Title:	Interface Consistency (Phase 2): Risk Assessment, Standards, and Guidelines		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
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Flight Program:			
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Key Personnel Changes/Previous PI:	Dr. Ryan Lange was added to the team in November, 2024, to aid with prototyping.		
COI Name (Institution):	Holden, Kritina Ph.D. (Leidos)		
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Task Description:	Future exploration missions will rely on the design and development of vehicles and complex systems from within NASA and through multiple external commercial partners to meet mission goals. Although consistency-related requirements exist, NASA's current approach to commercial spaceflight development is to allow providers great flexibility and opportunities to innovate. The degree of standardization across Artemis vehicle interfaces is a concern, as a lack of consistency (interface characteristics, interactivity) could impact crew performance and mission success. Results from the Interface Consistency Phase I pilot study suggest inconsistent interfaces are associated with higher workload, increased inefficiency (slower performance time, more actions required), and reduced satisfaction using the system. Determining the level of risk is difficult, as there is no information about the differential impact of specific types of inconsistency related to complex systems. These effects may be exacerbated under the high stress and levels of uncertainty associated with more autonomous future missions. The project team's previous research, Phase I: Characterization and Measurement, addressed many of the shortcomings found in the literature by creating a definition for consistency, formulating a taxonomy of tangible interface characteristics, applying this framework as guidance for developing consistency assessment tools, and conducting an exploratory pilot study to assess performance of these tools. Three prototypes were created for use in the pilot study, collectively exhibiting a range of interface characteristics representative of what crew may use during an Artemis mission and reflecting either high or low consistency with one another. Three different assessment tools were created and used to evaluate differences in consistency between the prototypes. Tools were developed to target different participant groups: the Checklist and Cognitive Walkthrough tools were designed to be used by user interface experts, and the Intersystem Consistency Scale (ICS) was designed to be used by end-user participants with crew-like demographics. This approach provides tool options for use early in the development process (when experts can identify issues for modification), as well as later in the process (when user testing is being performed). Preliminary results from the Phase I pilot study indicate that scores on each of the three assessment tools are able to discriminate between prototype interfaces demonstrating high or low global consistency with a reference prototype and that the tools were judged to be "easy to use". Results also suggest that performance differences, in general, can be observed between consistent and inconsistent interfaces. However, additional work planned for Phase II will: • Refine the assessment tools using feedback and results obtained during Phase I, including: item reduction, item phrasing, adding item descriptions and/or free-form responses, and weighting items or scales, if applicable • Characterize risk by understanding differences in human performance and subjective feedback related to each specific type of inconsistency (i.e., interface characteristics and interaction) • Understand whether this risk is mitigated with practice • Propose a set of standards and guidelines addressing aspects of inconsistent design that appear to have the most impact on performance
Rationale for HRP Directed Research:	Future crewed exploration missions beyond low Earth orbit (LEO), including Artemis, will be made possible by many development efforts from both within NASA and by external partners. The number of different vehicles and systems requiring human interaction will be much greater for future missions. Design solutions across programs and projects are likely to vary, and with NASA's current approach of allowing commercial providers great flexibility in design, the risk of inconsistency is significant. Artemis development efforts are already in progress, which underscores the urgency of this work. This proposal qualifies for directed study due to the insufficient time required for solicitation. Another factor is the highly constrained nature of the work. The proposers are currently involved in commercial programs as subject matter experts, and are very knowledgeable about the diversity of designs currently existing, and the development processes and constraints under which consistency measures and standards would have to be implemented. External researchers would not easily have access to this information; thus, directed research would be more cost effective. This work is leveraging the foundational work completed by the same project team, Interface Consistency Phase I: Characterization and Measurement.
Research Impact/Earth Benefits:	This research aims to better understand human performance impacts that are associated with externally inconsistent interfaces for complex systems. Determining the level of risk is difficult to estimate when reviewing the body of research literature, as there are no "gold standard" tools for assessing the level of consistency between systems and there is no information about the differential impact of specific types of inconsistency related to complex systems. The results could begin to describe this risk and methods that could be used to examine these impacts with complex ground systems and user populations with similar characteristics.
	Aim 1: Refine assessment tools using feedback and results obtained during Phase I, including: item reduction, item phrasing, adding item descriptions and/or free-form responses, and weighting items or scales, if applicable. In addition, provide practitioner documentation. Progress includes: o Selected the Checklist as the expert evaluator assessment tool. This includes refinement of checklist categories, wording, addition of a global consistency scale, and modifying the evaluation process by adding annotation. The tool has been tested with human-computer interaction experts at NASA's Ames Research Center. Results are in the works. A practitioner's guide for the Checklist is also in the works, which will provide guidance on applying the tool and interpreting its results. o The Intersystem Consistency Scale, the assessment tool developed for the crew, was further refined using results obtained during Phase I of the project. This included the reduction of redundant items and adding a "severity scale" to allow raters to describe the perceived performance impacts of the most salient inconsistencies. A crew interview is scheduled to evaluate these modifications. Once the scale is finalized, a practitioner guide will be created. Aim 2: Refine tasks and prototype stimuli using feedback from Program human interface and Artemis training Subject Matter Experts. Interviews were conducted with Training and Program Interface Subject Matter Experts (SMEs) to derive a more realistic set of tasks and interface attributes that may be expected for our exploration crews (i.e., across multiple vehicles within a single mission). These interviews were summarized in a report, delivered during the fall of 2024. The insights were used to create a final set of test procedures and to identify interface and interaction attributes that would be included as manipulations across prototypes. The prototype manipulations during Phase I included multiple dimensions

Task Progress:	of consistency (e.g., both interface and interactivity), but the prototypes for Phase II are concentrated on high/low levels of consistency for each dimension. For instance, of the five prototypes, one prototype is used as a baseline design from which all consistency judgments will be made. One will be highly consistent with this baseline design in both interface and interactivity. One will be highly consistent in interface, but inconsistent in interactivity. One will be highly consistent in interactivity, but inconsistent in interface attributes. The last will be inconsistent on both the interface and interactivity dimensions. The prototypes will be coded in Axure, capturing both time on task and click frequency. Three prototypes are currently finished, one is in the works, and one has yet to be started. The usability of each prototype will be independently tested with help from a university collaboration. Results will provide guidance as to whether particular designs should be further modified to achieve equitable usability across all prototypes. Aim 3: Characterize risk by understanding differences in human performance and subjective feedback related to each specific type of inconsistency (i.e., interface characteristics and interaction), low-level interface attributes, and the impact of switching between designs. A risk assessment study is planned to provide insight as to when inconsistency matters, using the tasks and prototypes discussed in Aim 2. A protocol has been developed and reviewed by the Human Factors and Behavioral Performance (HFBP) management. There will be four conditions. Within each condition, participants will be exposed to the baseline design and one additional design. Approximately 60 participants (approximately 15 participants for each of 4 conditions) will be recruited from the Johnson Space Center (JSC) Human Test Subjects Facility. There will be two study sessions: one for training the designs and one for testing the impact on performance related to switching between designs. During the first session, participants will receive training and complete several procedure trials for both the baseline design and a second design (one of the four other prototypes, which vary in level of consistency and consistency dimension). During the second session, participants will switch between the two designs and performance measures will be recorded. A dry run is expected mid-summer with data collection beginning mid-to-late summer 2025. Aim 4: Propose a set of standards and guidelines addressing aspects of inconsistent design that appear to have the most impact on performance. A set of standards and guidelines from industry, government, and agency sources was collected during Phase I of the project. Once results from the risk assessment study are in hand, this set will be further downselected and/or modified for potential inclusion into the NASA Standard 3001 Volume 2 and/or other applicable agency documents.
Bibliography Type:	Description: (Last Updated: 04/28/2026)
Conference Materials (Downloadable)	Smith A, Holden K, Robertson I, Karasinski J, Upasani S, Monsonyi J, Wu S-C, Parisi M, McTigue K, Lange R. "Interface consistency: Phase I results & phase II status." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 Interface Consistency Updates_IWS 2025_FINAL.pdf